# Technology Architecture

## Table of contents


Purpose of this chapter . . . 1
Position in the framework . . . 1
How NSIs use this chapter . . . 2
Technology layers . . . 2
Nodes . . . 2
System Software . . . 3
Technology Services . . . 4
Artifacts . . . 6


### Purpose of this chapter

The technology architecture describes the platform on which the applications run and on which the data is stored. It covers the nodes, the system software, the technology services that the platform offers to the applications, and the artifacts in which data objects take physical form.

Like the application components, the system software is named with concrete open-source products. The technology services are the stable part of the reference. An NSI can realise the same service with a different product, or obtain it from a shared or hosted platform.

### Position in the framework

In the TOGAF Architecture Development Method, the technology architecture is developed in Phase D. In ArchiMate, it is the technology layer, the lowest of the three core layers. It receives its requirements from the layers above and serves them.

**PLACEHOLDER: Position of the Technology Architecture below the Data and Application Architecture**

| Direction | Relationship | Example in the SAF |
|---|---|---|
| From Motivation | Requirements are realised by technology elements | SAF-RQ07 is realised by the five storage zones; SAF-RQ02 by the Event Streaming Service |
| From Strategy | Technology services realise capabilities | The Object Storage Service realises IT infrastructure (3.2.5) |
| From Data Architecture | Artifacts realise data objects | The Statistics Zone realises *Statistics Dataset - SDMX (#4)* |
| From Application Architecture | Technology services serve application components | The Relational Database Service serves eleven application components |

| Direction | Relationship | Example in the SAF |
|---|---|---|
| Within the layer | Nodes host system software; system software realises technology services | Storage Nodes host Ceph RGW, which realises the Object Storage Service |

## How NSIs use this chapter

An NSI uses this chapter as a checklist of the technology services that the application landscape needs, and as one worked example of how to provide them with open-source software.

The reference platform is a Kubernetes cluster that can run on-premise or in a hybrid setup. An NSI that uses a public or government cloud can obtain several technology services as managed services, for example object storage, relational databases or secrets management. The technology service in the SAF stays the same; only the element that realises it changes. As in the application architecture, licences are listed, and SAF-CN01 applies.

VS-09 Infrastructure and Platform Enablement describes the business view of providing these platforms and services. This chapter describes what is provided.

## Technology layers

The technology architecture contains 3 nodes, 15 system software elements, 12 technology services and 8 artifacts. They form four layers: nodes host system software, system software realises technology services, technology services serve the application components, and artifacts hold the data and the definitions.

**PLACEHOLDER: ArchiMate view *Phase D - Technology Architecture: Layers***

## Nodes

A **node** is a computational or physical resource that hosts other technology elements.

| ID | Node | Description | Composed of |
|---|---|---|---|
| **SAF-TN01** | Kubernetes Platform Cluster | Container platform that hosts all application components. On-premise or hybrid, with separate namespaces for production, test and research (secure) workloads. | SAF-TN02, SAF-TN03; SAF-SW02, SAF-SW11, SAF-SW12, SAF-SW13, SAF-SW14, SAF-SW15 |
| **SAF-TN02** | Compute Worker Nodes | General-purpose servers that run containers, Trino workers and statistical jobs. | SAF-SW01, SAF-SW03, SAF-SW05, SAF-SW06, SAF-SW07, SAF-SW08, SAF-SW09, SAF-SW10 |
| **SAF-TN03** | Storage Nodes | Servers with local disks that form the object-storage cluster. | SAF-SW04 |

## System Software

**System software** is software that provides an environment for storing, running and using other software or data.

| ID | System software | Licence | Description | Hosted on | Realises technology service |
|---|---|---|---|---|---|
| **SAF-SW01** | Linux | GPL-2.0 | Operating system for all nodes. | SAF-TN02 | |
| **SAF-SW02** | Kubernetes | Apache-2.0 | Container orchestration. Airflow KubernetesExecutor and Trino run on it. | SAF-TN01 | SAF-TSV03 |
| **SAF-SW03** | containerd | Apache-2.0 | Container runtime. | SAF-TN02 | SAF-TSV03 |
| **SAF-SW04** | S3-Compatible Object Storage (Ceph RGW) | LGPL-2.1 | Durable S3 API storage for all data zones. Alternative: MinIO (AGPL-3.0). | SAF-TN03 | SAF-TSV01 |
| **SAF-SW05** | Apache Iceberg Table Format | Apache-2.0 | Open table format on S3: ACID, schema evolution, snapshots, branches and tags (used as steady-state quality versions). | SAF-TN02 | SAF-TSV02 |
| **SAF-SW06** | Apache Kafka | Apache-2.0 | Event streaming backbone for registration events and lineage. | SAF-TN02 | SAF-TSV04 |
| **SAF-SW07** | PostgreSQL | PostgreSQL License | Relational store for Airflow, catalogs, Keycloak, .Stat Suite and the service desk. | SAF-TN02 | SAF-TSV05 |
| **SAF-SW08** | OpenSearch | Apache-2.0 | Search and index for the metadata catalogue and log search. | SAF-TN02 | SAF-TSV06 |

| ID | System software | Licence | Description | Hosted on | Realises technology service |
|---|---|---|---|---|---|
| **SAF-SW09** | OpenBao | MPL-2.0 | Secrets, certificates and transit encryption keys (used for pseudonymisation). Open-source fork of HashiCorp Vault. | SAF-TN02 | SAF-TSV07 |
| **SAF-SW10** | Observability Stack (Prometheus, OpenTelemetry, Grafana, Loki) | Apache-2.0 / AGPL-3.0 | Metrics, traces, logs and alerting for the platform and the pipelines. | SAF-TN02 | SAF-TSV08 |
| **SAF-SW11** | Harbor | Apache-2.0 | Container image registry with scanning and signing. | SAF-TN01 | SAF-TSV09 |
| **SAF-SW12** | Argo CD | Apache-2.0 | GitOps delivery of platform and application releases. | SAF-TN01 | SAF-TSV09 |
| **SAF-SW13** | OpenTofu | MPL-2.0 | Infrastructure as code for clusters, storage and networks. | SAF-TN01 | SAF-TSV11 |
| **SAF-SW14** | Velero | Apache-2.0 | Backup and restore of cluster state and volumes. | SAF-TN01 | SAF-TSV10 |
| **SAF-SW15** | Traefik | MIT | Ingress controller with TLS termination. | SAF-TN01 | SAF-TSV12 |

## Technology Services

A **technology service** is the behaviour that the platform offers to application components and to other technology services.

| ID | Technology service | Description | Realised by | Serves | Realises |
|---|---|---|---|---|---|
| **SAF-TSV01** | Object Storage Service | S3 API for the landing, raw, standardized, processed, statistics and released zones. | SAF-SW04 | SAF-AC02, SAF-AC03, SAF-AC05, SAF-AC08, SAF-AC17; SAF-TSV02 | 3.2.5 IT infrastructure |

| ID | Technology service | Description | Realised by | Serves | Realises |
|---|---|---|---|---|---|
| **SAF-TSV02** | Table Storage Service | Iceberg tables with snapshots, branches and tags on object storage. | SAF-SW05 | SAF-AC05, SAF-AC06, SAF-AC07, SAF-AC08 | 3.2.5 IT infrastructure |
| **SAF-TSV03** | Container Orchestration Service | Scheduling, scaling and isolation of containers; jobs for statistical workloads. | SAF-SW02, SAF-SW03 | SAF-AC04, SAF-AC05, SAF-AC08, SAF-AC17 | 3.2.5 IT infrastructure |
| **SAF-TSV04** | Event Streaming Service | Durable events for *Registered* signals and lineage. | SAF-SW06 | SAF-AC03, SAF-AC04, SAF-AC11 | SAF-RQ02 |
| **SAF-TSV05** | Relational Database Service | Managed PostgreSQL instances. | SAF-SW07 | SAF-AC04, SAF-AC06, SAF-AC11, SAF-AC12, SAF-AC13, SAF-AC15, SAF-AC16, SAF-AC18, SAF-AC21, SAF-AC22, SAF-AC23 | |
| **SAF-TSV06** | Search and Index Service | Search over metadata and logs. | SAF-SW08 | SAF-AC11, SAF-AC16 | |
| **SAF-TSV07** | Secrets and Key Management Service | Secrets, certificates and transit keys. | SAF-SW09 | SAF-AC03, SAF-AC04, SAF-AC05, SAF-AC18 | 3.3.6 IT security services |
| **SAF-TSV08** | Observability Service | Metrics, logs, traces and alerts. | SAF-SW10 | SAF-AC04, SAF-AC05, SAF-AC20 | 3.2.3 Technical management |
| **SAF-TSV09** | Image Registry and Delivery Service | Trusted images; together with Argo CD it delivers all releases (GitOps). | SAF-SW11, SAF-SW12 | SAF-TSV03 | |
| **SAF-TSV10** | Backup and Recovery Service | Backup of cluster state and persistent volumes. | SAF-SW14 | SAF-TSV01, SAF-TSV05 | |
| **SAF-TSV11** | Infrastructure Provisioning Service | Reproducible infrastructure from code. | SAF-SW13 | SAF-TSV03 | |
| **SAF-TSV12** | Ingress and TLS Service | Secure external access to portals and APIs. | SAF-SW15 | SAF-AC13, SAF-AC14, SAF-AC15, SAF-AC16, SAF-AC17, SAF-AC20 | |

## Artifacts

An **artifact** is a piece of data that is used or produced by software, such as a file, a table or a repository. In the SAF, artifacts are the physical form of the data objects. All artifacts are stored on the S3-compatible object storage (SAF-SW04).

The five zones are the physical steady states. Each zone is an Iceberg namespace, and each registered version in it is an immutable, tagged snapshot (SAF-RQ07).

| ID | Artifact | Description | Realises |
|---|---|---|---|
| **SAF-AF01** | Raw Zone (S3 bucket + Iceberg namespace `raw`) | Landing prefix plus registered raw tables. Write-once; BIV classification tags; retention per agreement. | SAF-DO01, SAF-RQ07 |
| **SAF-AF02** | Standardized Zone (namespace `std`) | Pseudonymised standardized tables that conform to the classifications. | SAF-DO02, SAF-RQ07 |
| **SAF-AF03** | Processed Zone (namespace `proc`) | Edited and enriched microdata with quality-version tags and edit lineage. | SAF-DO03, SAF-RQ07 |
| **SAF-AF04** | Statistics Zone (namespace `stat`) | SDMX-shaped tables (dimension, measure and attribute columns per DSD) with accuracy attributes. Approved products as dataflows. | SAF-DO04, SAF-RQ07 |
| **SAF-AF05** | Released Zone (namespace `rel`) | Disclosure-controlled SDMX-shaped tables and the published SDMX messages. | SAF-DO06, SAF-RQ07 |
| **SAF-AF06** | SDMX Messages (SDMX-ML, SDMX-JSON, SDMX-CSV) | Files and API responses in SDMX formats, generated from the statistics and released zones. | SAF-DO04, SAF-DO06 |
| **SAF-AF07** | Airflow DAG Repository (Git) | DAG definitions, one set per stream, reviewed and released through GitLab. | SAF-DO18 |
| **SAF-AF08** | dbt Project (Git) | Versioned SQL models, tests and documentation per stream. | SAF-DO18 |